

ULC404: OPERATING SYSTEMS

L	T	P	Cr
3	0	2	4.0

Course Objectives: To understand the role, responsibilities, and algorithms involved for achieving various functionalities of an Operating System.

Syllabus

Introduction and Operating System Structures: Computer-System Organization, Computer-System Architecture, Operating-System Structure, Operating-System Operations, Computing Environments, Operating-System Services, User and Operating-System Interface, System Calls, Types of System Calls, System Programs, Operating-System Structure, System boot.

Process Management: Process Concept, Process Scheduling, Operations on Processes, Inter-process Communication, Overview of Threads, Multi-core Programming, Multithreading Models, CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Multiple-Processor Scheduling.

Deadlocks: System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock.

Memory Management: Basic Hardware, Address Binding, Logical and Physical Address, Dynamic linking and loading, Shared Libraries, Swapping, Contiguous Memory Allocation, Segmentation, Paging, Structure of the Page Table, Virtual Memory Management: Demand Paging, Page Replacement, Allocation of Frames, Thrashing.

Storage Management: Overview of Mass Storage Structure, Disk Structure, Disk Attachment, Disk Scheduling, RAID Structure; File Concept, Access Methods, Directory and Disk Structure, File-System Structure, File-System Implementation, Directory Implementation, Allocation Methods.

Protection and Security: Principles of Protection, Domain of Protection, Access Matrix, Implementation of the Access Matrix, The Security Problem, Program Threats.

Process Synchronization: The Critical-Section Problem, Peterson's Solution, Synchronization Hardware, Mutex Locks, Semaphores, Classic Problems of Synchronization, Overview of Monitors.

Laboratory Work Learn and practice basic Linux/Unix commands to create and manipulate files and directories; explore vi editor environment; Write C programs demonstrating fork (), exec (), wait (), sleep () functions in Linux/Unix platform; Write C/C++ programs for message passing , shared memory and CPU scheduling algorithms.

Course Learning Outcomes (CLOs) /Course Outcomes (COs):

On completion of this course, the students will be able to:

1. Describe the basics of an operating system, including the kernel, system calls, and computing environments.
2. Evaluate the effectiveness and trade-offs of different models of multithreading, scheduling algorithms, and methods for handling deadlocks, such as prevention, avoidance, detection, and recovery.
3. Understand components of a memory system, virtual memory and analyze different memory management techniques.
4. Evaluate the effectiveness of different Disk Management strategies, and Critique the design and implementation of File System
5. Explain the basic concepts of Concurrency, Protection and Security issues in an operating system.

Text Books:

1. Operating System Concepts, Silberschatz A., Galvin B. P. and Gagne G., John Wiley & Sons Inc., 9th ed, 2013.
2. Operating Systems Internals and Design Principles, Stallings W., Prentice Hall 9th ed, 2018

Reference Books:

1. Understanding the Linux Kernel, Bovet P. D., Cesati M., O'Reilly Media, 3rd ed, 2006.
2. Introduction to Operating System Design and Implementation: The OSP 2 Approach, Kifer M., Smolka A. S., Springer, 2007

Evaluation Scheme	
Evaluation Elements	Weightage %
Mid Semester Test (MST) and End Semester Examination (ESE)	70
Sessional may include assignments, regular lab assessments, quizzes, and minor projects (report, presentation etc.)	30